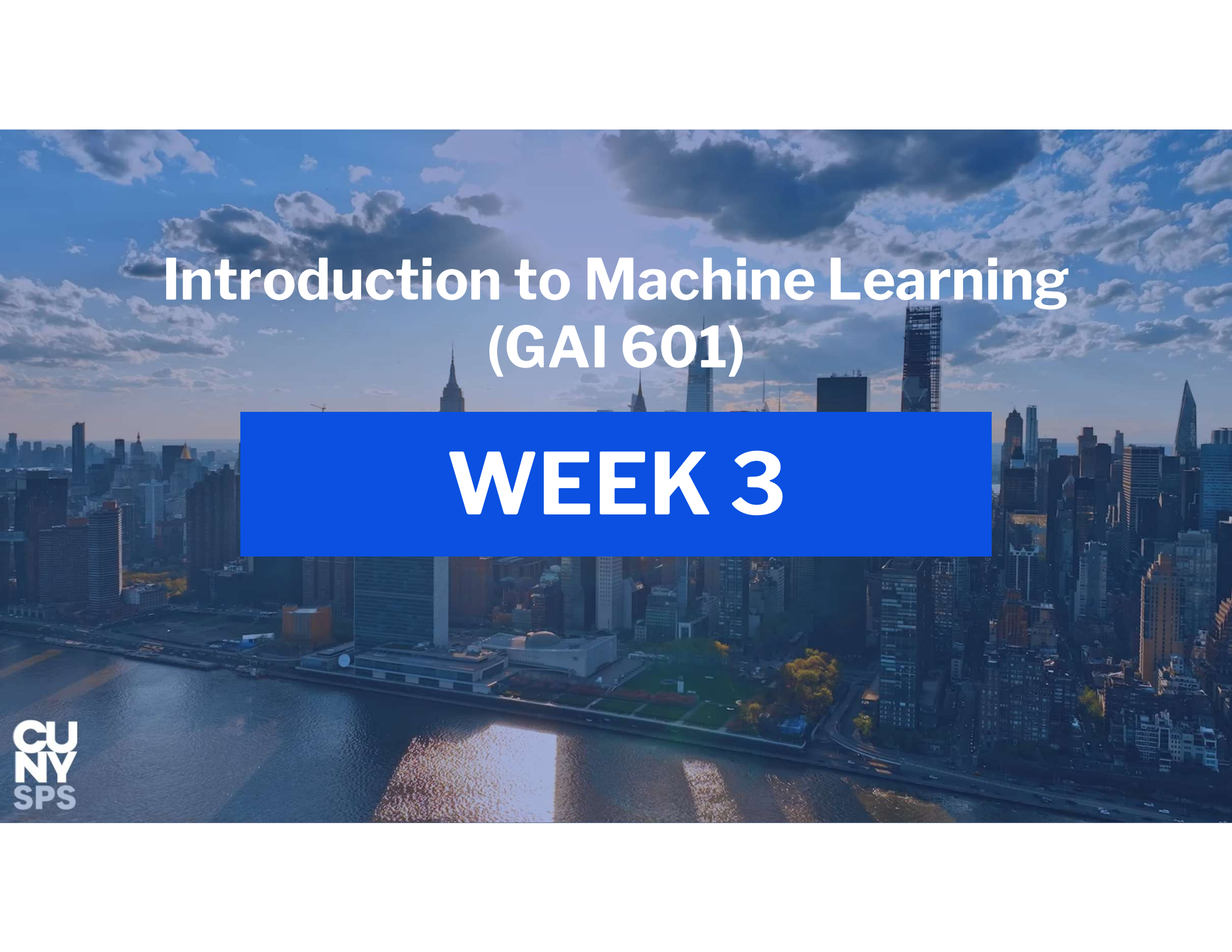
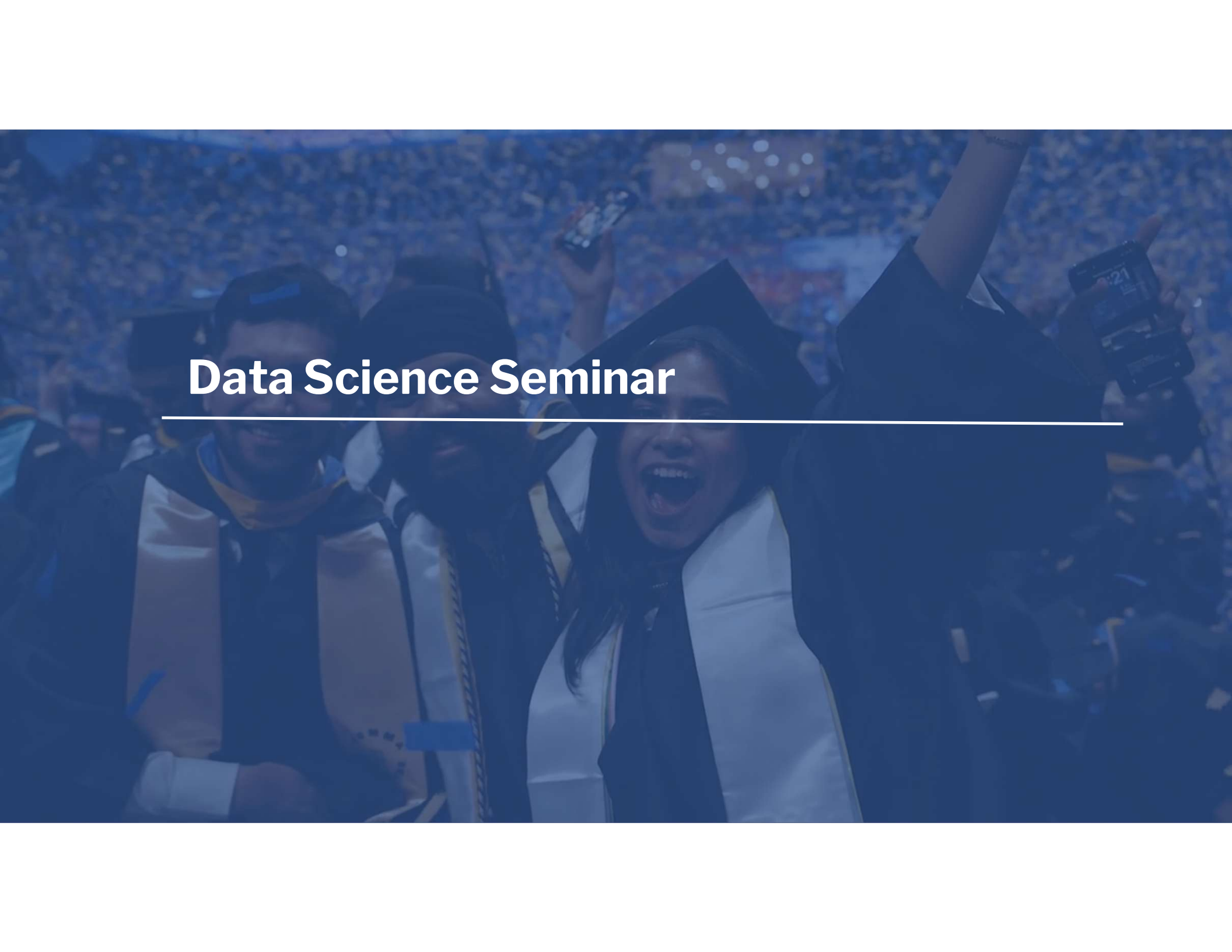


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large, semi-transparent blue rectangular box is centered over the image, containing white text. The sky is filled with scattered white clouds.

Introduction to Machine Learning (GAI 601)

WEEK 3

A photograph of graduates in black gowns and white stoles, celebrating with their arms raised and holding up smartphones. The scene is filled with blue and white confetti, creating a festive atmosphere. The image is overlaid with a semi-transparent blue filter.

Data Science Seminar

CUNY SPS Data Science Seminar

Dr. Adam Kelleher
Columbia University



Understanding and Using Causal Models for Forecasting

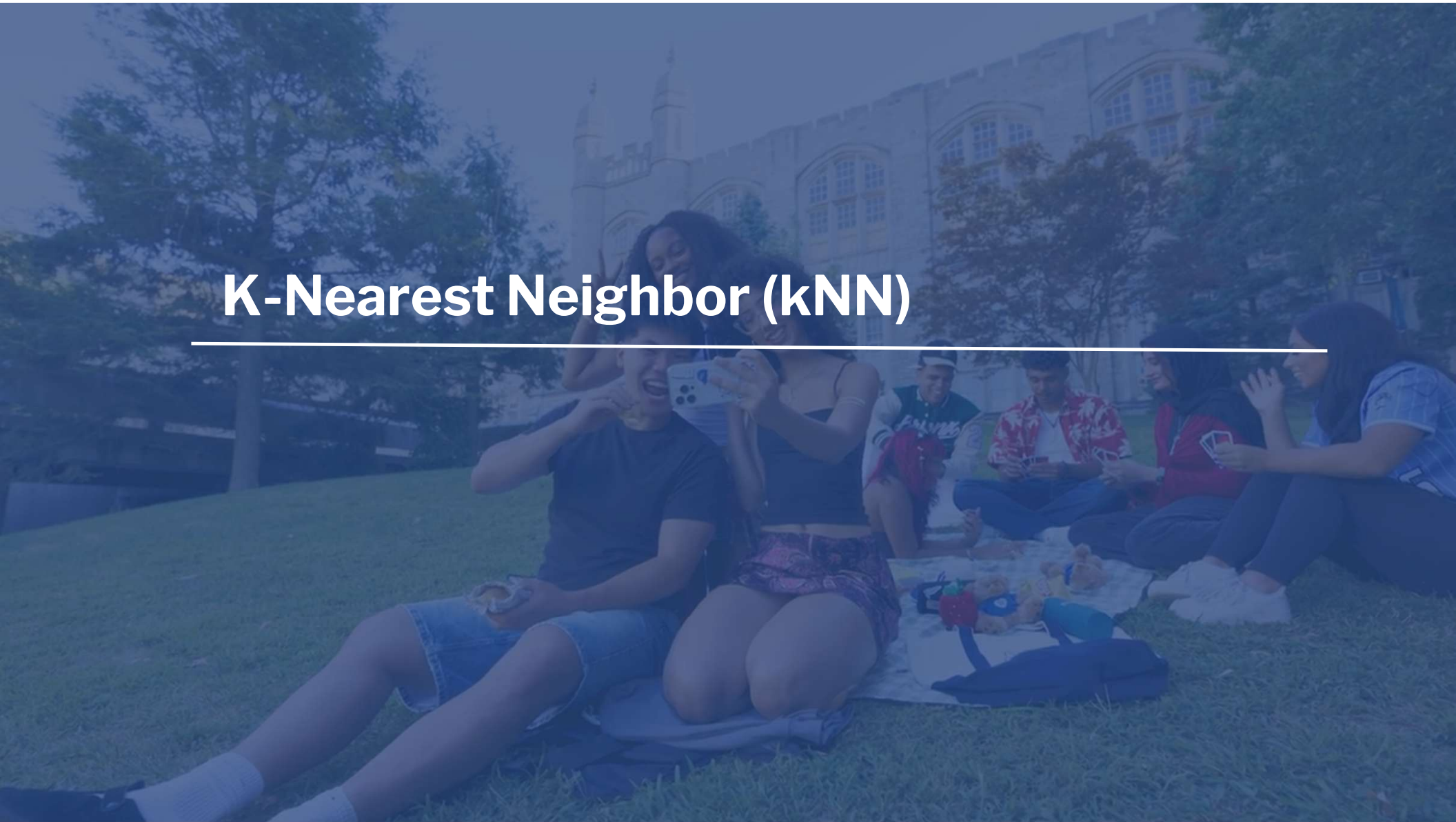
Causal Inference tells us what will happen if we make an intervention in a real world system. We will introduce the basic idea of Causality and discuss how to train Causal Models. Then we will discuss several real world case studies from a variety of different industries where using a Causal Model instead of a predictive one made a large difference to operations.

Thursday, October 1st
6:30PM
119th West 31st Street
Room 405

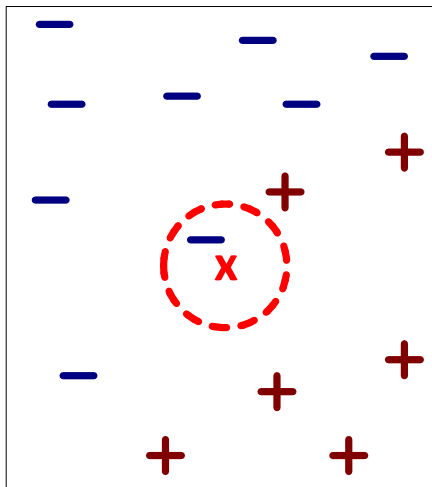
Sign up link (please sign up even if you are coming in person, you can put whatever you want for your emplid and the form will take it)
<https://us02web.zoom.us/j/83840080608?pwd=KwzeAlG66KJQ27QeAqBGkqkr5Xsxml.1&jst=1>



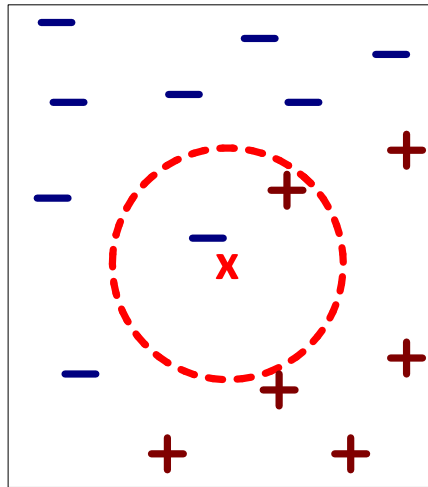
K-Nearest Neighbor (kNN)



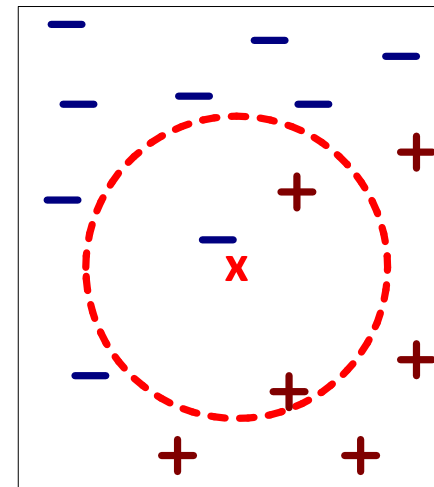
K-Nearest Neighbor (kNN)



(a) 1-nearest neighbor



(b) 2-nearest neighbor



(c) 3-nearest neighbor

Source: Supervised Learning Algorithms: A Comparison, November 2020 Kristu Jayanti Journal of Computational Sciences (KJCS) - DOI:10.59176/kjcs.v1i1.1259. Free access

Identifying k

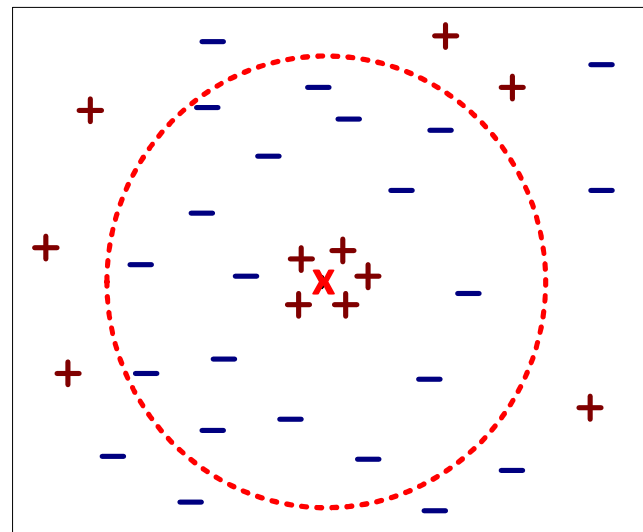
Choosing the value of k:

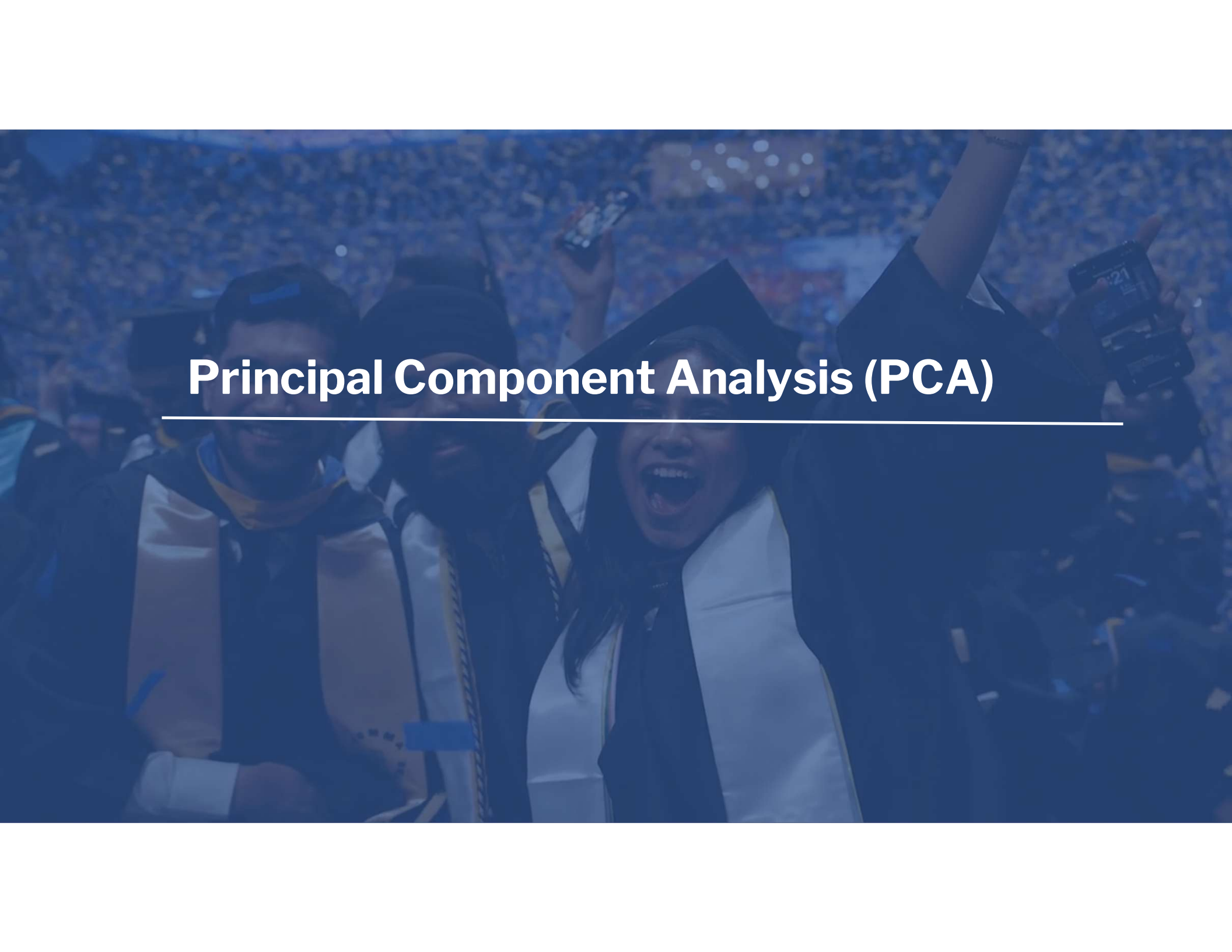
- If k is too small, sensitive to noise points
- If k is too large, neighborhood may include points from other

Rule of thumb:

$$k = \sqrt{N}$$

N: number of training points



A photograph of graduates in black gowns and white stoles, celebrating with confetti in the air. One graduate in the foreground is cheering with her mouth open. The image has a blue tint.

Principal Component Analysis (PCA)

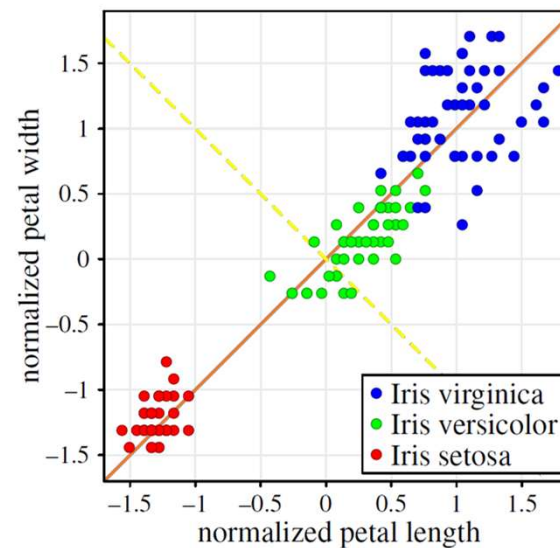
Principal Component Analysis (PCA)

- **Dimensionality reduction technique**
- **Project data from the high-dimensional space to a lower-dimensional space**
- **Criteria: Maximize data variance to construct principal components**

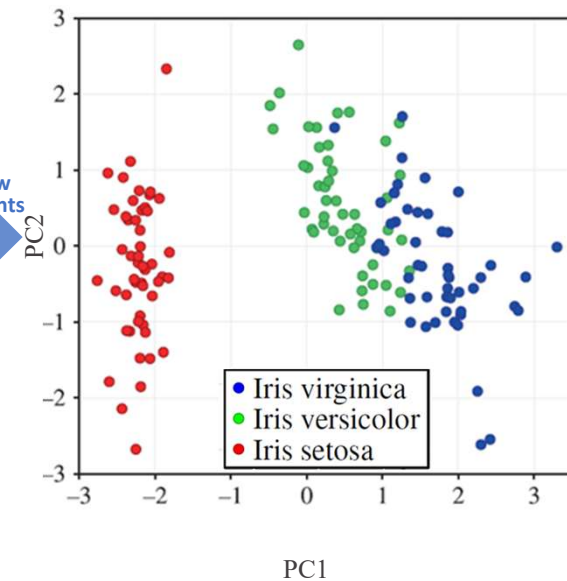
Steps:

1. Construct a line in the direction of maximum variance in data (PC1 = orange line)
2. Next component is orthogonal to the previous component (PC2 = yellow line)

Repeat for as many PC as you need.



Replot
onto new
components



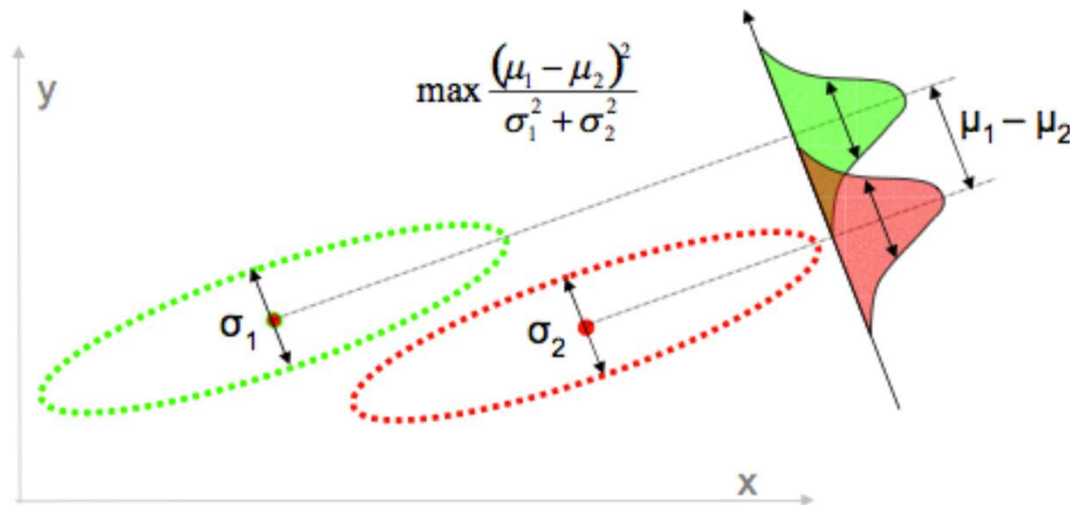
Source: Guide to Intelligent Data Science, Berthold et al

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed upper section and a clock face is visible. To the right, a modern building with a glass facade is partially obscured by trees. In the foreground, a wide, paved walkway leads towards the buildings, with several students walking along it. Some students are carrying backpacks. On the left, there are large, leafy trees and a black lamppost. A person is sitting on a bench in the lower-left corner. The overall scene is a typical university setting.

Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA)

- Dimensionality reduction approach
- Two criteria are used by LDA to create a new axis:
 1. Maximize the distance between means of the two classes.
 2. Minimize the variation (spread) within each class.



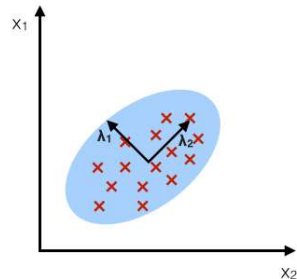
Source: Victor Lavrenko

Linear Discriminant Analysis (LDA)

- **Discriminants (LDA)** maximize the separation $_$ classes
- **Components (PCA)** maximize the variance in the data
- **Dimensionality reduction technique**
- **Project data from the high-dimensional space to a lower-dimensional space**
- **Criteria: Maximize data variance to construct principal components**

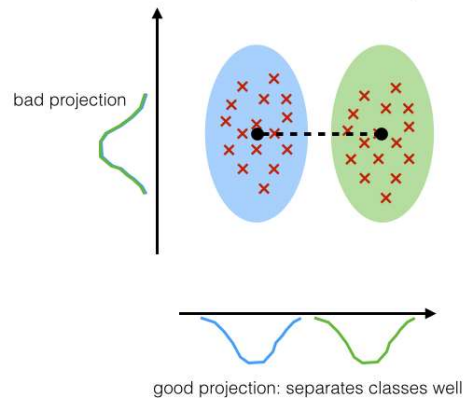
PCA:

component axes that maximize the variance



LDA:

maximizing the component axes for class-separation



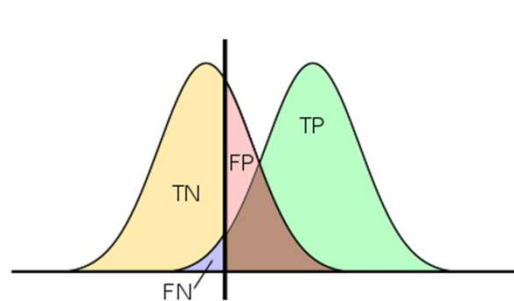
Source: Victor Lavrenko

Classification Metrics



Confusion Matrices

- **Accuracy:** Accuracy is the proportion of all classifications that were correct, whether positive or negative
- **Recall:** Recall is a measure of how many positives your model is able to recall from the data.
- **Precision:** Precision is the ratio of correct positive predictions to the total positive predictions.
- **F1 Score:** F1 score metric is used when you seek a balance between precision and recall.



ACTUAL VALUES	POSITIVE	NEGATIVE
POSITIVE	TP	FN
NEGATIVE	FP	TN

$$Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

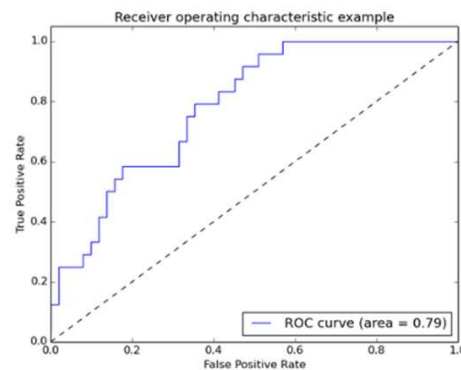
$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Source: Alleviating Class-Imbalance Data of Semiconductor Equipment Anomaly Detection Study,

ROC & AUC

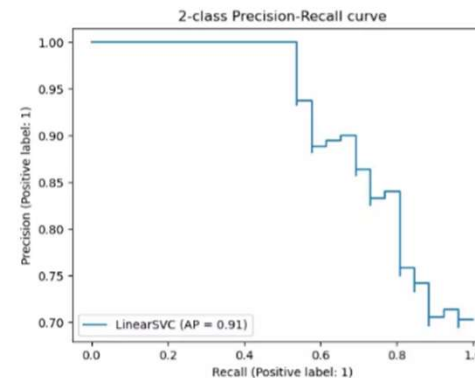
- **Area under the Curve (AUC):** The AUC represents the probability that the model, if given a randomly chosen positive and negative example, will rank the positive higher than the negative.
- **Receiver-operating characteristic curve (ROC):** A ROC curve is constructed by plotting the true positive rate (TPR) against the false positive rate (FPR). The ROC curve is drawn by calculating the true positive rate (TPR) and false positive rate (FPR) and then graphing TPR over FPR.

Receiver-operating characteristic curve (ROC):



Both Precision-Recall Curve and ROC-AUC curve are used:

- To explain model goodness of fit
- To identify the correct threshold to map probabilities value to the actual classes 0/1



When to use which one:

- Precision Recall curve is used when there is imbalance class distribution.
- ROC-AUC curve is used when there is balanced class distribution in data.

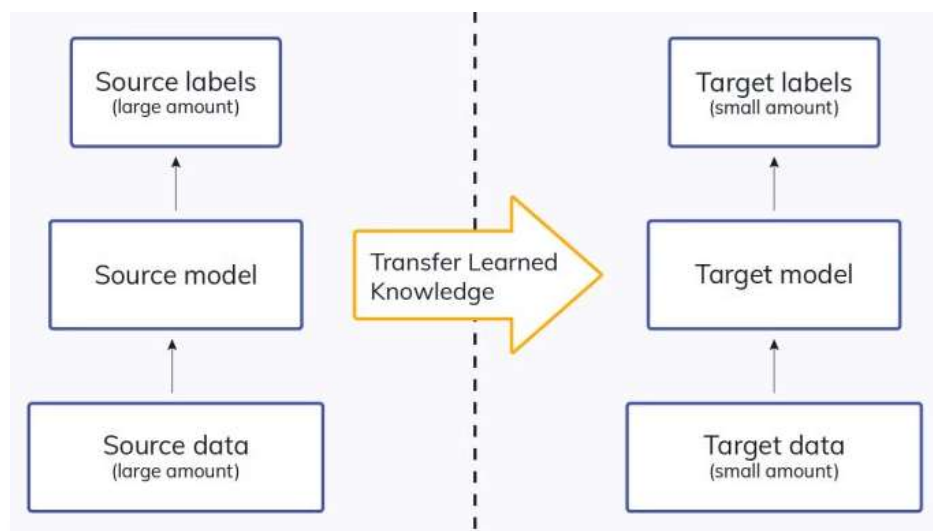
Source: <https://ashutoshtripathy.com>

Transfer Learning



Transfer Learning

- **Transfer learning** is a machine learning technique where a model developed for one task is reused as the starting point for a model on a second task. It allows training to be “jump-started”.
- For example, a model trained for self-driving for cars may be useful for self-driving trucks
- **Benefits:**
 1. Reduced data requirements
 2. Faster training
 3. Reduce compute & lower costs

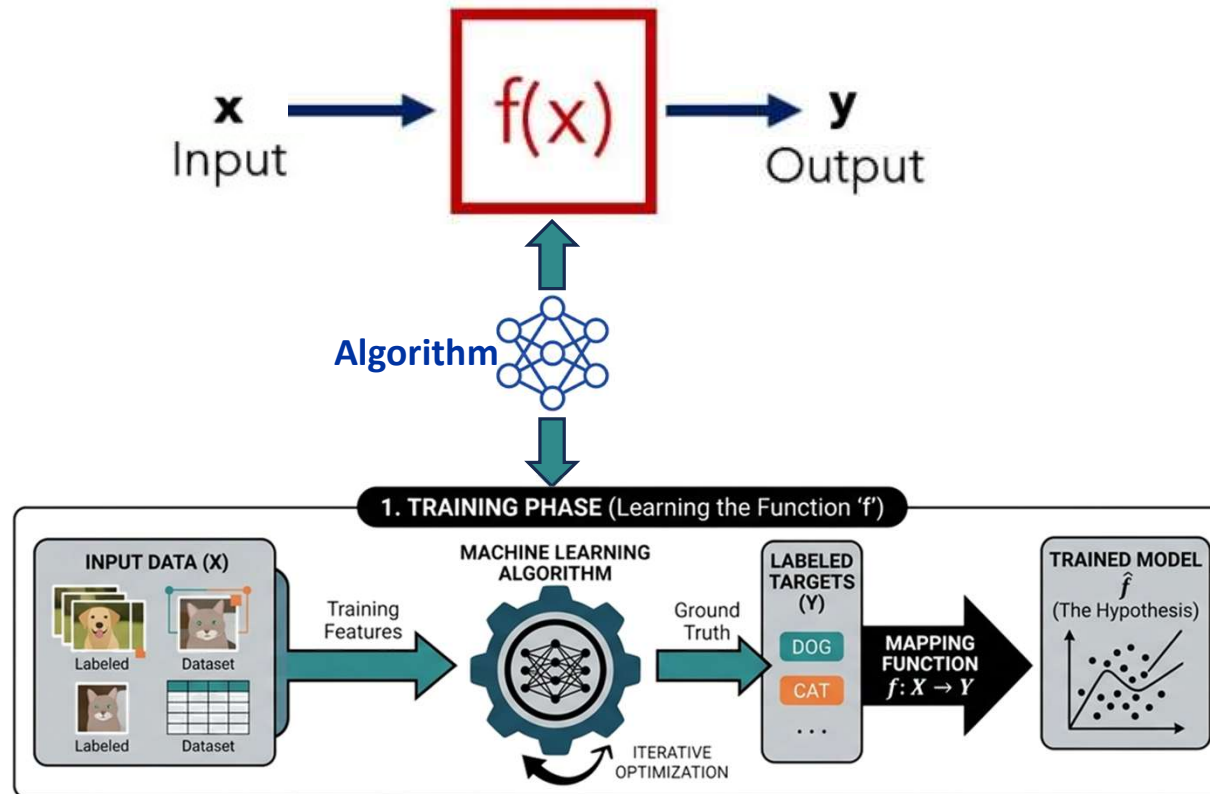


Source: v7labs

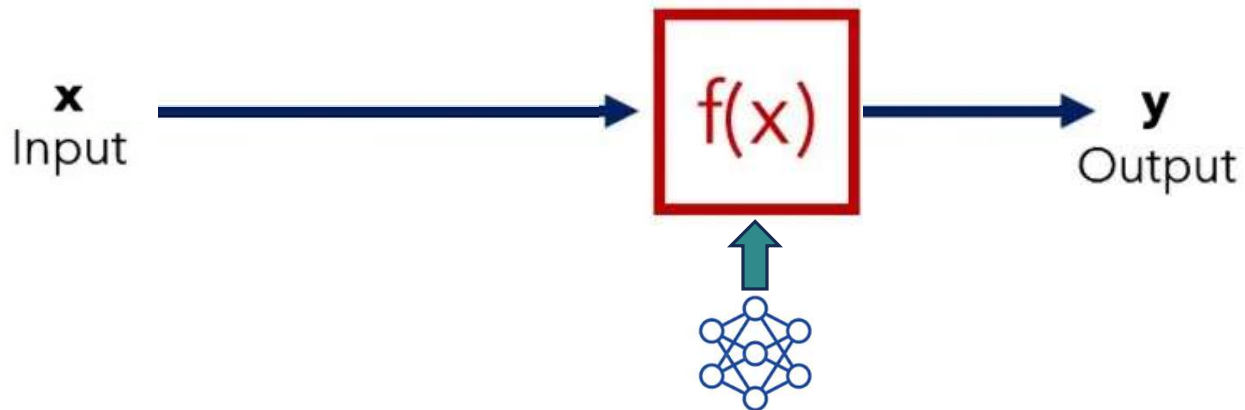
A photograph of a university campus scene, featuring a prominent clock tower in the background, surrounded by trees and a paved walkway with students. The image is overlaid with a semi-transparent blue filter.

Business Context (Assignment 1)

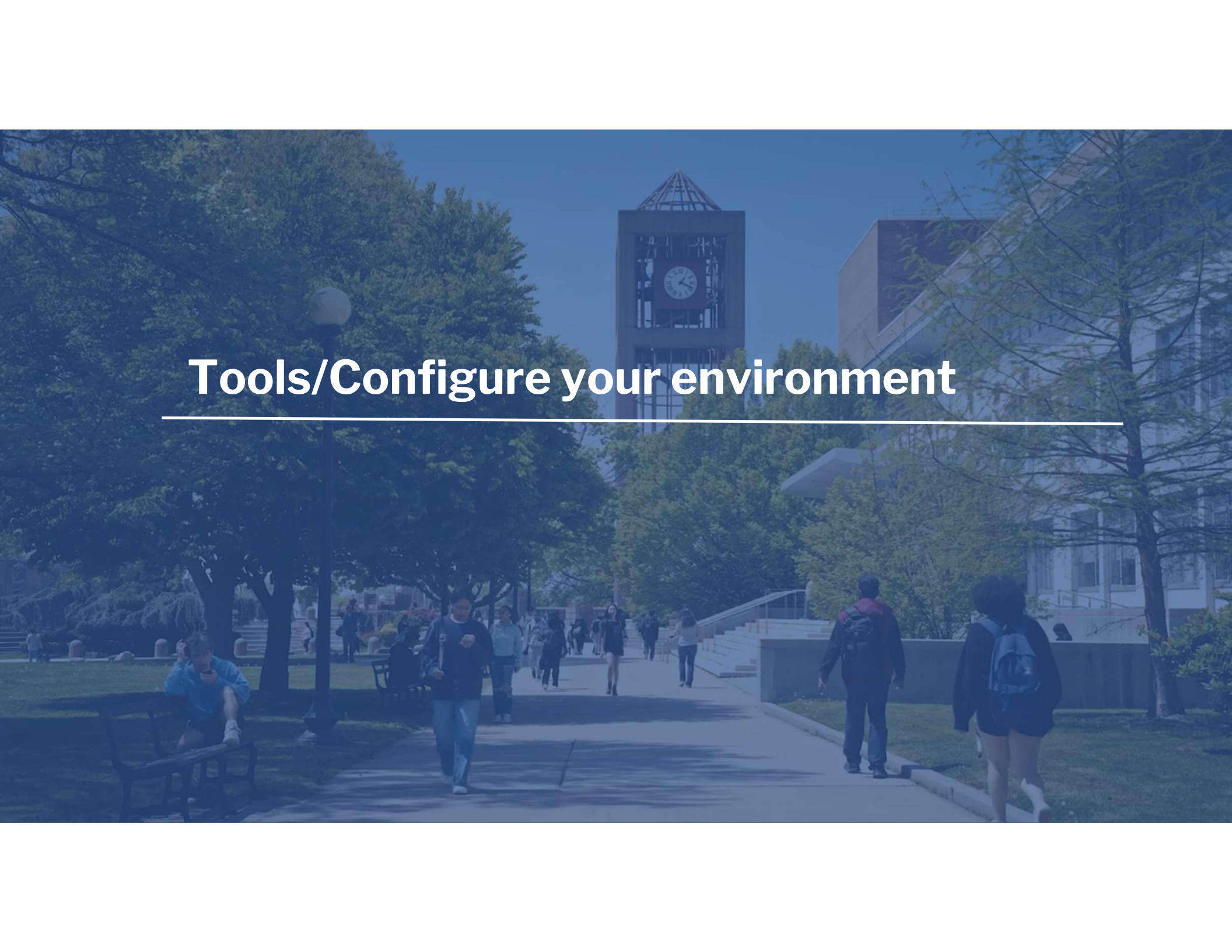
ML as a Function



The business problem



Use Case	X (Input)	Algorithm	Y (Output)
1 What's the problem you want to solve?	2 What data do you need?	3 Which system and approach do you want to use?	4 What do you want to Predict

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground shows a wide, paved pedestrian walkway lined with mature trees. Several students are walking along the path; some are carrying backpacks. On the left, a person is sitting on a wooden bench. The overall atmosphere is that of a busy academic environment.

Tools/Configure your environment

Register for free tools

1. Colab

- Go to <https://colab.research.google.com/signup>
- Create a Google account (if you don't have one)
- Sign up for **Colab Pro for Education** (free for students)

2. GitHub Copilot

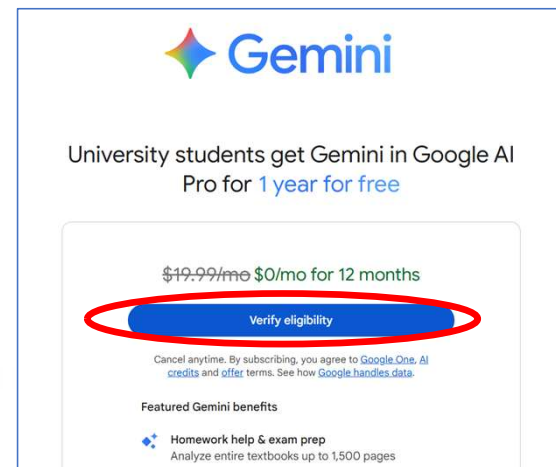
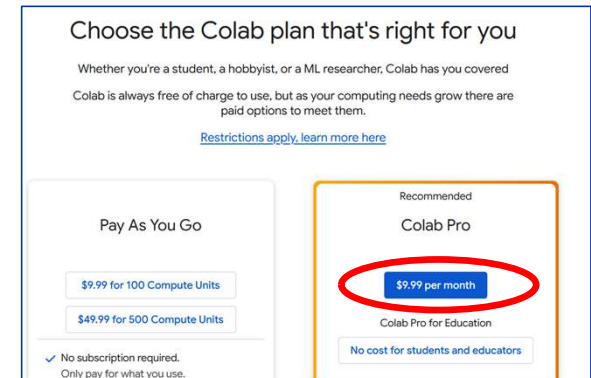
- Go to <https://github.com/education>
- Click on **Join GitHub Education**
- Create a GitHub account (if you don't have one)
- Click on Start an Application to get access

3. Google Gemini

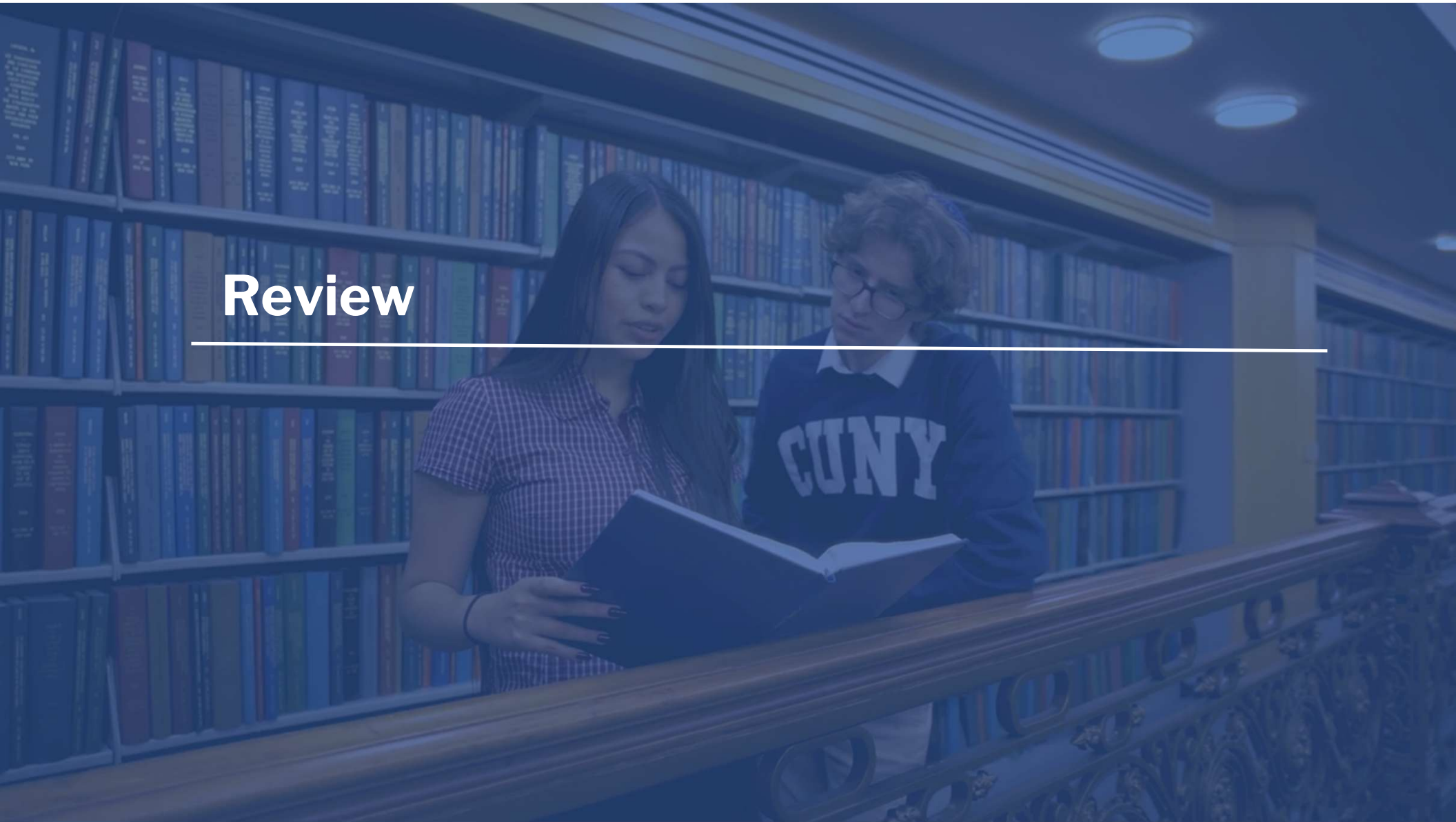
- Go to <https://gemini.google/us/students>
- Fill out the verification form

4. Microsoft Copilot 365

- Free with you Office 365 access (via CUNY)



Review



This week we covered

Lesson Objectives/Topics

1. Apply discriminant analysis and k-NN to classify observations in structured datasets.
2. Explain when to choose k-NN over other classification methods in business problems.
3. Prepare and normalize data for non-parametric classification models.
4. Compare classifier accuracy using confusion matrices and cross-validation.

Appendix



Training Data: Iris data set

Features (inputs)					Labels (target outputs)
Instance	Sepal Length (cm)	Sepal width (cm)	Petal length (cm)	Petal width (cm)	Class
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
...					
50	7	3.2	4.7	1.4	Iris-versicolor
51	6.4	3.2	4.5	1.5	Iris-versicolor
52	6.9	3.1	4.9	1.5	Iris-versicolor
53	5.5	2.3	4	1.3	Iris-versicolor
54	6.5	2.8	4.6	1.5	Iris-versicolor
55	5.7	2.8	4.5	1.3	Iris-versicolor
56	6.3	3.3	4.7	1.6	Iris-versicolor
...					
100	6.3	3.3	6	2.5	Iris-virginica
101	5.8	2.7	5.1	1.9	Iris-virginica
102	7.1	3	5.9	2.1	Iris-virginica
103	6.3	2.9	5.6	1.8	Iris-virginica
104	6.5	3	5.8	2.2	Iris-virginica
105	7.6	3	6.6	2.1	Iris-virginica



Iris Setosa

y_1



Iris Versicolor

y_2



Iris Virginica

y_3

There are 4 features (inputs): x_1, x_2, x_3 & x_4

There are 3 potential labels (outputs): y_1, y_2 , & y_3

Source: Iris data set, Fisher, et al

Label Space

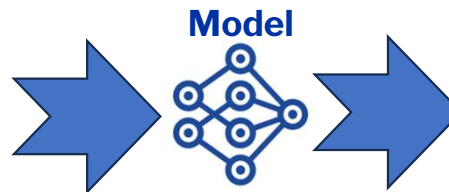
The trained model will map the Feature space to the Label (target output) space:

Feature space (inputs)

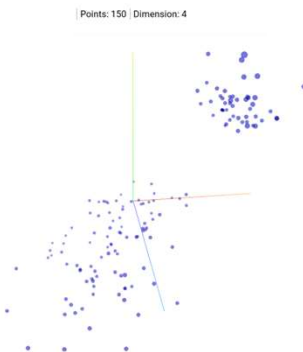
Sepal Length (cm)	Sepal width (cm)	Petal length (cm)	Petal width (cm)
5.1	3.5	1.4	0.2
4.9	3	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5	3.6	1.4	0.2
5.4	3.9	1.7	0.4
...			
7	3.2	4.7	1.4
6.4	3.2	4.5	1.5
6.9	3.1	4.9	1.5
5.5	2.3	4	1.3
6.5	2.8	4.6	1.5
5.7	2.8	4.5	1.3
6.3	3.3	4.7	1.6
...			
6.3	3.3	6	2.5
5.8	2.7	5.1	1.9
7.1	3	5.9	2.1
6.3	2.9	5.6	1.8
6.5	3	5.8	2.2
7.6	3	6.6	2.1

Labels (target outputs)

Class
Iris-setosa
Iris-setosa
Iris-setosa
Iris-setosa
Iris-setosa
Iris-setosa
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-versicolor
Iris-virginica
Iris-virginica
Iris-virginica
Iris-virginica
Iris-virginica
Iris-virginica



What the shape of the features looks like:



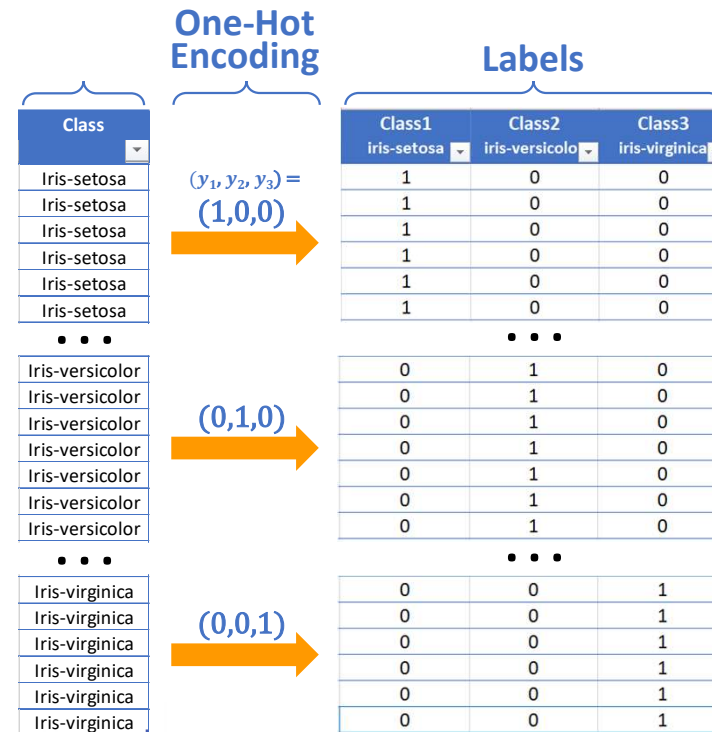
What does the label (target output) space look like?



Source: Iris data set, Fisher, et al

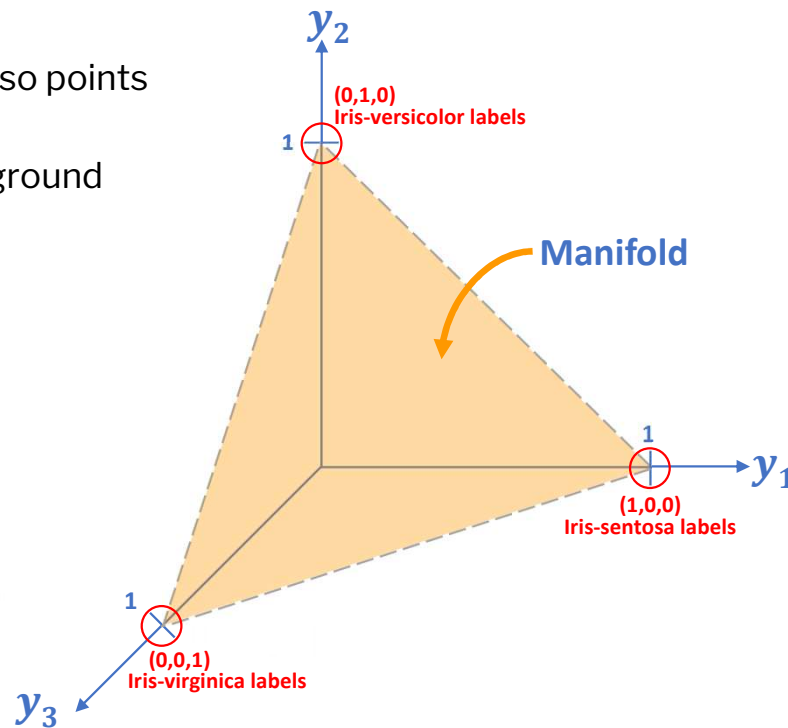
One-hot encoding

- ML requires numbers: labels must be converted to numbers
- Each class (type of label must be its own dimension)
- The value in each dimension conveys the probability it is of that class
- Training Data Labels always have a probability of 1 (100%) i.e. they are the “Ground Truth”



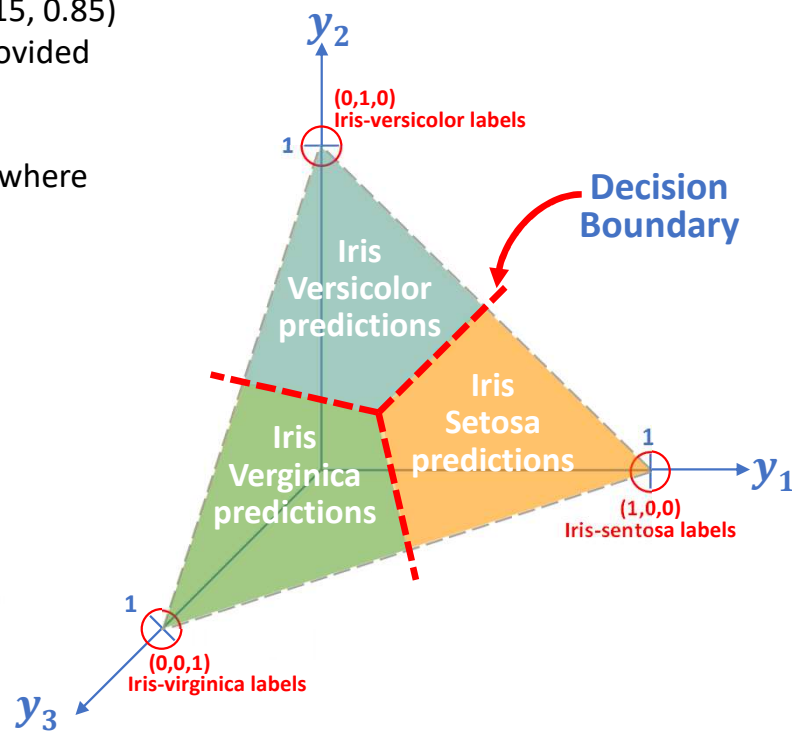
One-hot encoding

- The **number of dimensions = number of classes**.
In this case 3 dimensions.
- A Label (or prediction) is one data-point in that 3-dimensional space
- Probabilities of all classes add up to 1 (100%) so points **all points must lie on a manifold**
- Only labels have values of 1 (as they are the “ground truth”)



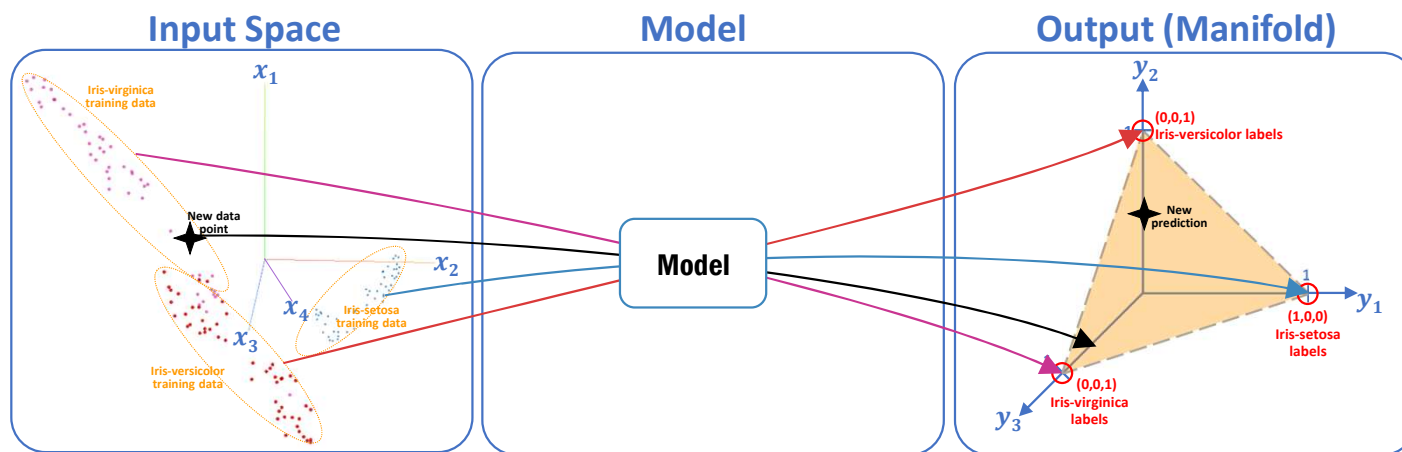
Decision Boundary

- The **prediction generates a probability for each class**, which must add to 1 (100%)
- For example, a prediction that $(y_1, y_2, y_3)=(0.1, 0.215, 0.85)$ means that the model predicts that the features provided have a 85% probability of being Iris Virginica.
- A **decision boundary separates the classes** – it lies where there is equal probability between classes.



Putting it all together

A predictive AI model is trained to map the Feature Space to the Target (Label) space.



An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's steel structure and suspension cables are prominent. The water of the harbor is visible in the lower portion of the frame. The sky is a clear, pale blue.

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